University of the Western Cape
Universiteit van Wes-Kaapland

COS 711—Machine Learning
Gradient descent and linear regression.
Test 1 Term 1 **9 March 2015**

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1. (a) Who gave this definition of machine learning: "A computer program is said to *learn* from *experience* E with respect to some *task* T and some *performance measure* P if its performance on T, as measured by P, improves with experience E." [3]

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   (b) Suppose we feed a learning algorithm a lot of historical weather data, and have it learn to predict weather. Explain in terms of this task what each of T, E and P entail. [9]

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2. Give an example of a regression problem and a classification problem in machine learning. [2]

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3. Suppose you have a store of the genetic data for many thousands of people and you know the attributes of "the gene that causes diabetes". You have a program that can find the gene and thus predict that the individual has diabetes. What must you do to measure the accuracy or error rate of your program when running it on your training set? Do you need any data that

has not been mentioned? [3]

4. Consider the following arrays.

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \text{ and } \mathbf{x} = \begin{bmatrix} 5 \\ 7 \end{bmatrix} \text{ and } \mathbf{y} = \begin{bmatrix} 6 \\ 12 \end{bmatrix} \text{ and } \mathbf{ones} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Answer each of the following questions by giving the result of the operation or when asked give the OCTAVE that will generate the correct answer.

(a) Give the value of `A * B`. [3]

(b) Give the value of `A .* B`. [3]

(c) Give the value of `x' * y`. [3]

(d) Give the value of `x * y'`. [3]

(e) Give the value of `x .* y`. [3]

(f) Give OCTAVE to do the following: let $X \leftarrow B$, and add **ones** as the first column of $\mathbf{X}$, and then solve for $\mathbf{\Theta}$ using linear regression to solve the equations $\mathbf{X\Theta} = \mathbf{y}$. [3]

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andIn this assignment We want find a best fit for the hypothesis

$$h_\theta(x) = \theta_0 x_0^{(i)} + \theta_1 x_2^{(i)} + \ldots + \theta_n x_n^{(i)}$$

using linear regression and gradient descent.

We require *verbose* but sensible documentation—contrary to our usual requirements of overly concise documentation. Use sensible names for variables, but by all means use the names of standard statistical or mathematical variables such such as `standardDeviation` or better because it is shorter `sigma` for σ but `averageX` or better `meanX` for $\overline{x}$. Avoid using silly abbreviations such as `av` for the mean. Use matrix and vector computations as far as possible. Follow the steps in order.

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1. Read in the 5-D training data set `trainfeatures.dat`. Each line i for $i = 1..m$ has values for the five features $x_1^{(i)}, x_2^{(i)}, x_3^{(i)}, x_4^{(i)}, x_5^{(i)}$. The labels are given the data set `trainlabels.dat`. The labels correspond to $y^{(i)}$. Column 1 contains the x_1-values, and column 2 the x_1-values, ... [1]
2. Construct each the five vectors $x^{(i)}$ and $y^{(i)}$ and plot each of the five xs against y in a single figure. Save the figure as `figure1.eps`. [2]
3. Construct a matrix X by adding the column of constant values 1 (for 0) to x.
4. Compute the solution using the normal equation. [3]
5. Your submission The program should contain (and call) the following functions (implemented in .m files):
   (a) `costFuncLinear(X, theta, y)` to compute the cost function $J(\theta)$ where `J`, `X` and `y` are the parameters defined in class. Return the value of `J`. [2]
   (b) `doGradDescentLinear`—choose the parameters to provide to the function and the result(s) returned by the function. [10]
   (c) Present your results for the training data set given (ass1.txt). Plot the function. [2]

Total [20]

Submission

Present your results of the regression problem in a LaTeXdocument stored in a subdirectory of this assignment called `docs` containing a `make clean` of the directory. Make a directory called `surname,firstname` and create another directory inside it called `11assignment`. Every assignment that you create must be done in its appropriate directory, e.g., the next assignment will be done in the directory `21assignment` that lies in parallel with `11assignment`. Run all the code for the current assignment inside its directory and do not delete older work. After you have completed all the instructions for a given assignment create a tarball by using

```
tar -zcvf surname,firstname.tgz surname,firstname
```

and then mail `surname,firstname.tgz` to `uwcmachinelearning@gmail.com`. You will be marked on your submissions to this mailbox. Late submissions will not be tolerated.

Hints

1. Use the `load` command and not `fopen`, `textscan`, `fclose`.
2. Generalize programs by using `length` or `size` of vectors and matrices instead hard-coding the number of items per dimension.
3. Initialize θ with `zeros`, `ones`, `rand`, `randn`) to start off.
4. Vectorize the the calculation of the cost function $J(\theta)$. Do not use the classical non-vectorized algorithm.
5. Spell check you `assignment11.tex` using `aspell -c assignment11.tex`. We only accept reports produced in LaTeX.
6. Programs with syntax errors or missing functions will be awarded 0.
7. Do Internet research and do not forget to reference it fully by citing the the source.
8. You are expected be be able to explain your code.
9. Please read the instructions carefully and follow them exactly.
